

$\mathcal{K}_0^{\#1} \dagger$
 $\mathcal{K}_0^{\#1} \dagger^{\alpha\beta\chi}$

$\mathcal{K}_0^{\#1}$
 $\mathcal{K}_0^{\#1}{}_{\alpha\beta\chi}$

$\frac{1}{2}(-Y_1 k^2 - 3 \overset{(2)}{\mathcal{K}}_3)$	0
0	0

$\mathcal{K}_1^{\#1} \dagger^{\alpha\beta}$
 $\mathcal{K}_1^{\#2} \dagger^{\alpha\beta}$
 $\mathcal{K}_1^{\#1} \dagger^\alpha$
 $\mathcal{K}_1^{\#2} \dagger^\alpha$

$\frac{k^2 \overset{(4)}{\mathcal{K}}_4}{2}$	$\frac{k^2 \overset{(4)}{\mathcal{K}}_4}{2\sqrt{2}}$	0	0
$\frac{k^2 \overset{(4)}{\mathcal{K}}_4}{2\sqrt{2}}$	$\frac{k^2 \overset{(4)}{\mathcal{K}}_4}{4}$	0	0
0	0	$-k^2 \overset{(4)}{\mathcal{K}}_2 - \overset{(2)}{\mathcal{K}}_3$	$\frac{Y_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_3}{2\sqrt{2}}$
0	0	$\frac{Y_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_3}{2\sqrt{2}}$	$\frac{1}{4}(-Y_3 k^2 - 2 \overset{(2)}{\mathcal{K}}_3)$

$\mathcal{K}_2^{\#1} \dagger^{\alpha\beta}$
 $\mathcal{K}_2^{\#1} \dagger^{\alpha\beta\chi}$

$-\frac{Y_4 k^2}{2}$	0
0	0

$\mathcal{J}_0^{\#1} \dagger$
 $\mathcal{J}_0^{\#1} \dagger^{\alpha\beta\chi}$

$\mathcal{J}_0^{\#1}$
 $\mathcal{J}_0^{\#1}{}_{\alpha\beta\chi}$

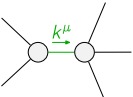
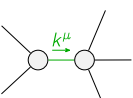
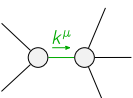
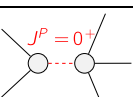
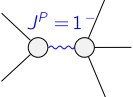
 $\mathcal{J}_1^{\#1} \dagger^{\alpha\beta}$
 $\mathcal{J}_1^{\#2} \dagger^{\alpha\beta}$
 $\mathcal{J}_1^{\#1} \dagger^\alpha$
 $\mathcal{J}_1^{\#2} \dagger^\alpha$
 $\mathcal{J}_2^{\#1} \dagger^{\alpha\beta}$
 $\mathcal{J}_2^{\#1} \dagger^{\alpha\beta\chi}$

Abbreviations used in matrices

$$Y_1 == 3 \overset{(4)}{\mathcal{K}}_1 + 3 \overset{(4)}{\mathcal{K}}_2 + 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4 \ \&\& \ Y_2 == 2 \overset{(4)}{\mathcal{K}}_2 + \overset{(4)}{\mathcal{K}}_7 \ \&\& \ Y_3 == 2 \overset{(4)}{\mathcal{K}}_2 + 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4 + 2 \overset{(4)}{\mathcal{K}}_7 \ \&\& \\ Y_4 == 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4 \ \&\& \ \text{Det}(0^+) == \frac{1}{2}(-k^2(3 \overset{(4)}{\mathcal{K}}_1 + 3 \overset{(4)}{\mathcal{K}}_2 + 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) - 3 \overset{(2)}{\mathcal{K}}_3) \ \&\& \\ \text{Det}(1^+) == \frac{1}{8}(k^4(2 \overset{(4)}{\mathcal{K}}_2(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) - \overset{(4)}{\mathcal{K}}_7^2) + 2 k^2(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) \overset{(2)}{\mathcal{K}}_3) \ \&\& \ \text{Det}(2^+) == -\frac{1}{2} k^2(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4)$$

Lagrangian

$$\overset{(2)}{\mathcal{K}}_3 \mathcal{K}^\alpha{}_\beta \mathcal{K}^\chi{}_{\beta\chi} + \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{\mathcal{K}}_2 \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\alpha\chi} + \overset{(4)}{\mathcal{K}}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} + \\ \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} + \overset{(4)}{\mathcal{K}}_4 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} + \overset{(4)}{\mathcal{K}}_7 \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}^\delta{}_{\chi\beta} + \frac{1}{4} \overset{(4)}{\mathcal{K}}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_1^{\#1\alpha\beta} == \mathcal{J}_1^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_\delta$	
Total # constraint(s):	9		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\overset{(4)}{\mathcal{K}}_4}$
	1	0	$\frac{1}{\overset{(4)}{\mathcal{K}}_4(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4)^2 \overset{(2)}{\mathcal{K}}_3} (-6 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4 \overset{(2)}{\mathcal{K}}_3 - 3 \overset{(4)}{\mathcal{K}}_4^2 \overset{(2)}{\mathcal{K}}_3 + 16 \overset{(4)}{\mathcal{K}}_3^2 \overset{(4)}{\mathcal{K}}_4 p^2 +$ $16 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4^2 p^2 + 4 \overset{(4)}{\mathcal{K}}_4^3 p^2 + 36 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7 p^2 + 18 \overset{(4)}{\mathcal{K}}_4^2 \overset{(4)}{\mathcal{K}}_7 p^2 + 18 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7^2 p^2 -$ $2 \sqrt{(\overset{(4)}{\mathcal{K}}_4^2 (208 \overset{(4)}{\mathcal{K}}_3^4 p^4 + 13 \overset{(4)}{\mathcal{K}}_4^4 p^4 + 81 \overset{(4)}{\mathcal{K}}_7^4 p^4 - 15 \overset{(4)}{\mathcal{K}}_4^3 p^2 (\overset{(2)}{\mathcal{K}}_3 - 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 27 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7^2 p^2 (-\overset{(2)}{\mathcal{K}}_3 + 6 \overset{(4)}{\mathcal{K}}_7 p^2) +$ $18 \overset{(4)}{\mathcal{K}}_4^2 (7 \overset{(2)}{\mathcal{K}}_3^2 - 3 \overset{(4)}{\mathcal{K}}_7 \overset{(2)}{\mathcal{K}}_3 p^2 + 11 \overset{(4)}{\mathcal{K}}_7^2 p^4) - 8 \overset{(4)}{\mathcal{K}}_3^3 (15 \overset{(2)}{\mathcal{K}}_3 p^2 - 2 (26 \overset{(4)}{\mathcal{K}}_4 + 45 \overset{(4)}{\mathcal{K}}_7) p^4) +$ $12 \overset{(4)}{\mathcal{K}}_3^2 (42 \overset{(2)}{\mathcal{K}}_3^2 - 3 (5 \overset{(4)}{\mathcal{K}}_4 + 6 \overset{(4)}{\mathcal{K}}_7) \overset{(2)}{\mathcal{K}}_3 p^2 + 2 (13 \overset{(4)}{\mathcal{K}}_4^2 + 45 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7 + 33 \overset{(4)}{\mathcal{K}}_7^2) p^4) + 2 \overset{(4)}{\mathcal{K}}_3 (52 \overset{(4)}{\mathcal{K}}_4^3 p^4 -$ $45 \overset{(4)}{\mathcal{K}}_4^2 p^2 (\overset{(2)}{\mathcal{K}}_3 - 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 27 \overset{(4)}{\mathcal{K}}_7^2 p^2 (-\overset{(2)}{\mathcal{K}}_3 + 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 36 \overset{(4)}{\mathcal{K}}_4 (7 \overset{(2)}{\mathcal{K}}_3^2 - 3 \overset{(4)}{\mathcal{K}}_7 \overset{(2)}{\mathcal{K}}_3 p^2 + 11 \overset{(4)}{\mathcal{K}}_7^2 p^4))))$
	1	0	$\frac{1}{\overset{(4)}{\mathcal{K}}_4(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4)^2 \overset{(2)}{\mathcal{K}}_3} (-6 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4 \overset{(2)}{\mathcal{K}}_3 - 3 \overset{(4)}{\mathcal{K}}_4^2 \overset{(2)}{\mathcal{K}}_3 + 16 \overset{(4)}{\mathcal{K}}_3^2 \overset{(4)}{\mathcal{K}}_4 p^2 +$ $16 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4^2 p^2 + 4 \overset{(4)}{\mathcal{K}}_4^3 p^2 + 36 \overset{(4)}{\mathcal{K}}_3 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7 p^2 + 18 \overset{(4)}{\mathcal{K}}_4^2 \overset{(4)}{\mathcal{K}}_7 p^2 + 18 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7^2 p^2 +$ $2 \sqrt{(\overset{(4)}{\mathcal{K}}_4^2 (208 \overset{(4)}{\mathcal{K}}_3^4 p^4 + 13 \overset{(4)}{\mathcal{K}}_4^4 p^4 + 81 \overset{(4)}{\mathcal{K}}_7^4 p^4 - 15 \overset{(4)}{\mathcal{K}}_4^3 p^2 (\overset{(2)}{\mathcal{K}}_3 - 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 27 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7^2 p^2 (-\overset{(2)}{\mathcal{K}}_3 + 6 \overset{(4)}{\mathcal{K}}_7 p^2) +$ $18 \overset{(4)}{\mathcal{K}}_4^2 (7 \overset{(2)}{\mathcal{K}}_3^2 - 3 \overset{(4)}{\mathcal{K}}_7 \overset{(2)}{\mathcal{K}}_3 p^2 + 11 \overset{(4)}{\mathcal{K}}_7^2 p^4) - 8 \overset{(4)}{\mathcal{K}}_3^3 (15 \overset{(2)}{\mathcal{K}}_3 p^2 - 2 (26 \overset{(4)}{\mathcal{K}}_4 + 45 \overset{(4)}{\mathcal{K}}_7) p^4) +$ $12 \overset{(4)}{\mathcal{K}}_3^2 (42 \overset{(2)}{\mathcal{K}}_3^2 - 3 (5 \overset{(4)}{\mathcal{K}}_4 + 6 \overset{(4)}{\mathcal{K}}_7) \overset{(2)}{\mathcal{K}}_3 p^2 + 2 (13 \overset{(4)}{\mathcal{K}}_4^2 + 45 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7 + 33 \overset{(4)}{\mathcal{K}}_7^2) p^4) + 2 \overset{(4)}{\mathcal{K}}_3 (52 \overset{(4)}{\mathcal{K}}_4^3 p^4 -$ $45 \overset{(4)}{\mathcal{K}}_4^2 p^2 (\overset{(2)}{\mathcal{K}}_3 - 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 27 \overset{(4)}{\mathcal{K}}_7^2 p^2 (-\overset{(2)}{\mathcal{K}}_3 + 6 \overset{(4)}{\mathcal{K}}_7 p^2) + 36 \overset{(4)}{\mathcal{K}}_4 (7 \overset{(2)}{\mathcal{K}}_3^2 - 3 \overset{(4)}{\mathcal{K}}_7 \overset{(2)}{\mathcal{K}}_3 p^2 + 11 \overset{(4)}{\mathcal{K}}_7^2 p^4))))$
	1	$-\frac{3 \overset{(2)}{\mathcal{K}}_3}{3 \overset{(4)}{\mathcal{K}}_1 + 3 \overset{(4)}{\mathcal{K}}_2 + 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4}$	$-\frac{2}{3 \overset{(4)}{\mathcal{K}}_1 + 3 \overset{(4)}{\mathcal{K}}_2 + 2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4}$
	3	$\frac{2(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) \overset{(2)}{\mathcal{K}}_3}{-2 \overset{(4)}{\mathcal{K}}_2 (2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) + \overset{(4)}{\mathcal{K}}_7^2}$	$\frac{2(4 \overset{(4)}{\mathcal{K}}_3^2 + \overset{(4)}{\mathcal{K}}_4^2 + 2 \overset{(4)}{\mathcal{K}}_4 \overset{(4)}{\mathcal{K}}_7 + 3 \overset{(4)}{\mathcal{K}}_7^2 + 4 \overset{(4)}{\mathcal{K}}_3 (\overset{(4)}{\mathcal{K}}_4 + \overset{(4)}{\mathcal{K}}_7))}{(2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) (2 \overset{(4)}{\mathcal{K}}_2 (2 \overset{(4)}{\mathcal{K}}_3 + \overset{(4)}{\mathcal{K}}_4) - \overset{(4)}{\mathcal{K}}_7^2)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		